

Gurinder Singh

Backend Engineer · Technical Trainer

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SUMMARY

Backend engineer and technical trainer with 3+ years delivering hands-on technical instruction alongside production software development. Build and operate my own infrastructure — three live services running on a self-managed Linux server behind Nginx with automated TLS and managed Postgres. Working with agentic AI systems: LLM API integration, tool and function calling, retrieval-augmented generation, and agent orchestration. Combine genuine engineering depth with proven curriculum design and one-to-one mentoring, including accessible instruction for screen reader users. Currently completing an MCA at VIT Bhopal University.

EXPERIENCE

Technical Trainer · Vision Aid

January 2023 – Present

- Deliver technical training in Advanced MS Excel, computer applications, and technical computing to differently-abled learners, with progression tracked through practical assessment.
- Design course curriculum and training material adapted for screen reader users — accessibility built into instructional design rather than retrofitted.
- Mentor learners one-to-one, adapting explanation and pacing to individual technical level and prior exposure.
- Contribute to the Learning & Development team on course structure, sequencing, and delivery method.

DSA Training & Internship · Skolar

May 2023 – July 2023

- Completed a three-month intensive programme in data structures and algorithms, focused on complexity analysis and optimisation.
- Implemented sorting, searching, graph traversal, and dynamic programming solutions against practical problem sets.

PROJECTS

CrisisCore — AI Crowd Safety & Emergency Response Platform

Team Lead

Java · Spring Boot · Spring Security · PyTorch · FastAPI · React · Kotlin · Docker · Nginx — crisiscore.gurindersingh.dev

- Led a team of five to build a crowd-safety and emergency-response platform for mass gatherings such as the Kumbh Mela, across eight modules; designed the architecture and the API contract the team built against in parallel. MCA major project, VIT Bhopal University.
- Built a Spring Boot risk engine that turns noisy CSNet crowd-density readings into Safe, Watch and Critical states using median smoothing, least-squares trend detection, hysteresis and dwell time — raising a critical alert 36 seconds before a simulated surge reached danger.
- Implemented SOS dispatch with scored unit selection, queuing and a strict incident state machine (median 9 ms from SOS to assigned unit), plus per-zone forecasting of resource shortfalls.
- Protected pilgrim data with AES-256-GCM field encryption, HMAC-hashed phone matching, recorded consent and audited contact reveal; secured staff access with JWT, role-based permissions and per-IP rate limiting.
- Verified with 72 unit tests and an 87-check end-to-end suite; deployed on Oracle Cloud with Docker Compose, Nginx and automated TLS, alongside a Kotlin Android app for one-tap SOS.

FPMIS — Forensic Post-Mortem Intelligence System

Lead Developer

FastAPI · Python · PostgreSQL · Next.js 14 · TypeScript · JWT · Docker · Nginx — fpmis.gurindersingh.dev

- Led a team of five developers to ship a full-stack forensic case management platform across eight integrated modules; selected as a finalist and presented at HealthHack 2026, VIT Bhopal University.
- Designed a 100-point weighted matching algorithm linking unidentified bodies to missing-person records across seven attributes, returning ranked results with confidence tiers.
- Built secure REST APIs with JWT authentication, role-based access control, and full audit logging over a normalised PostgreSQL schema.
- Deployed to production on a self-managed Linux server — containerised with Docker Compose behind an Nginx reverse proxy with automated certificate renewal.

Baazi Club — Accessible Android Card Game Platform

Solo Developer

Java · Android · Firebase · Realtime Database · WebRTC · TalkBack — baaziclub.net · in closed testing

- Built an accessibility-first Android card game platform shipping four games — Bhaabi, Seep, Ninety-Nine, and Hand Cricket — with full TalkBack support and spoken game state throughout.
- Implemented a shared game engine spanning four rulesets with real-time multiplayer state synchronisation over Firebase Realtime Database.
- Built WebRTC voice rooms holding up to fifteen players, persisting in the background during active play.

- Handled room lifecycle management including locking, master transfer, and automatic host promotion; Android App Links resolve invite URLs into the waiting room via a verified assetlinks contract.

Personal Platform & Infrastructure

Solo Developer

Java · Spring Boot · PostgreSQL · Flyway · Docker · Nginx · Oracle Cloud — api.gurindersingh.dev

- Built and deployed a Spring Boot REST API serving live project data and contact submissions to a public portfolio site, backed by managed PostgreSQL with Flyway-versioned migrations.
- Provisioned and administer the underlying Ubuntu ARM server on Oracle Cloud — three services behind an Nginx reverse proxy, each with its own auto-renewing TLS certificate.
- Kept application state entirely in managed Postgres so the host remains disposable and reproducible; frontend deploys continuously from Git.

SKILLS

Agentic AI & LLM Engineering: LLM API integration, prompt engineering, tool and function calling, retrieval-augmented generation (RAG), vector embeddings, agent orchestration and multi-step workflows, Model Context Protocol (MCP), evaluation and guardrails

Java Backend: Java, Spring Boot, Spring MVC, Spring Security, JWT, Hibernate, JPA, RESTful APIs, JUnit 5, Maven

Python Backend: Python, FastAPI, SQLAlchemy, Pydantic, RESTful APIs, PyTorch

Databases: PostgreSQL, MySQL, MongoDB, Firebase Realtime Database, Flyway

Infrastructure & DevOps: Docker, Docker Compose, Nginx, Linux (Ubuntu) administration, Oracle Cloud, AWS, TLS/Certbot, Git, continuous deployment

Frontend: TypeScript, Next.js, React, JavaScript, semantic HTML, WCAG AA

Training & Delivery: Curriculum development, technical instruction, accessible course design, one-to-one mentoring, assessment design, team leadership

TRAINING TOPICS DELIVERED

Agentic AI fundamentals and agent architectures · LLM API integration and prompt design · tool and function calling · RAG pipelines · backend API development with Java and Spring Boot · Python and FastAPI · relational database design · containerisation and deployment · accessible software engineering · advanced MS Excel and technical computing.

EDUCATION

Master of Computer Applications (MCA)

2025 – Present

VIT Bhopal University, Bhopal

Bachelor of Computer Applications (BCA)

Completed May 2025

Christ University, Bengaluru

CERTIFICATIONS & LANGUAGES

- AWS Cloud Foundations — Amazon Web Services
- Data Structures and Algorithms — Skolar
- Languages: English · Hindi · Punjabi